

Assignment - 25

What is meant by Array in Java

- Same as C & C++

- Syntactical diff in Java

- 2 major Categories $\leftarrow$ single dimensional
multidimensional

code : Array1.java & Array2.java

Can create jagged array in C? How?

m: array of arrays with diff sizes

- In C, jagged arrays cannot be created directly but can be simulated using pointers.

#include <stdio.h>

#include <stdlib.h>

int main () {

int * arr[3]; // array of integer pointers.

arr[0] = (int*) malloc (2 * sizeof(int));

arr[1] = (int*) ——— (3* ———);

arr[2] = ——— (4* ———);

arr[0][0] = 1; arr[0][1] = 2;

arr[1][0] = 3; arr[1][1] = 4; arr[1][2] = 5;

arr[2][0] = 6; arr[2][1] = 7; arr[2][2] = 8;

arr[2][3] = 9;

printf ("ddd", arr[2][3]);

return 0;

3] Jagged Array

Structure
Memory
Arrays of array with diff sizes
Efficient (only required space
is allocated)

Arrays with fixed sizes
Less efficient (fixed size)

Declaration
Req.
`int[][] arr = new int[3][5];`
`{ {1, 2}, {3, 4, 5}, {6} }`

`int[][] arr = new int[3][3];`
`{ {1, 2, 3}, {4, 5, 6}, {7, 8, 9} }`

Q1] What is 'length' property in Java?

- It returns the size of an array
- Not a method, but a property

Req.
`int[] arr = {10, 20, 30, 40, 50};`
`System.out.println(arr.length);` // output: 5
• For a 2D array, `arr.length` returns the no. of rows & `arr[row].length` returns the no. of columns

Q2] Explain Input mechanisms in Java.

- Java provides multiple ways to take user input:

1) Using Scanner

```
import java.util.Scanner;
Scanner sc = new Scanner(System.in);
int num = sc.nextInt();
```

2) Using BufferedReader

```
import java.util.Scanner;
import java.io.*;
BufferedReader br = new BufferedReader(new InputStreamReader(
    Reader System.in));
String str = br.readLine();
```

3) Using Console Class

```
String input = System.console().readLine();
```

Q1 Explain mechanism to accept an integer using Scanner class.

→ Create object of Scanner & use nextInt() method.

ex:
 import java.util.Scanner;
 public class InputExample {
 public static void main (String[] args) {
 Scanner sc = new Scanner (System.in);
 System.out.println ("Enter a No.");
 int num = sc.nextInt();
 System.out.println ("You entered: " + num);
 sc.close();
 }
 }
 // sc.nextInt(); reads an integer entered by user

Q2 Write a Java Program to accept N no.s from user, store them in an array & display largest no.

→ import java.util.Scanner;
 public class LargestNumber {
 public static void main (String[] args) {
 Scanner sc = new Scanner (System.in);
 System.out.println ("Enter no. of elements: ");
 int n = sc.nextInt();
 int[] arr = new int[n];
 System.out.println ("Enter " + n + " numbers:");
 for (int i = 0; i < n; i++)
 { arr[i] = sc.nextInt(); }
 int max = arr[0];
 for (int i = 1; i < n; i++) {
 if (arr[i] > max) { max = arr[i]; } }
 System.out.println ("Largest number: " + max);
 sc.close();
 }
 }

Q1) Write a Java Program to accept N no.s, store them in an array & display the avg.

```

→ import java.util.Scanner;

public class AvgNumber {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter no. of elements: ");
        int n = sc.nextInt();
        int [] arr = new int[n];

        System.out.println ("Enter " + n + " numbers:");
        int sum = 0;
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }

        double avg = (double) sum / n;
        System.out.println ("Average: " + avg);
        sc.close();
    }
}

```

Q2) Wrapper Class in Java?

- Primitive datatypes in Java have corresponding class for it (converted into objects)
- known as - wrapper class

Boxing - Converts primitive → corresponding W.C.
 int → Integer
 int → Boxing → Integer.

9/ Unboxing - Class obj $\rightarrow$ corresponds primitive datatype
 Integer $\rightarrow$ (Unboxing) $\rightarrow$ int
Code : Wrapper.java

10/ Java Program $\rightarrow$ Create a Jagged array,
 No. of rows & sizes of each row, accept from user.

```
import java.util.Scanner;
public class JaggedArray {
    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter no. of rows: ");
        int rows = sc.nextInt();
        int[][] jagged = new int[rows][];
```

1) Accept row sizes

```
for (int i = 0; i < rows; i++) {
    System.out.println("Enter size of row" + (i+1) + ": ");
    int cols = sc.nextInt();
    jagged[i] = new int[cols];
```

1) Accept elements

```
System.out.println("Enter elements: ");
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < jagged[i].length; j++) {
        jagged[i][j] = sc.nextInt();
    }
}
```

1) Display jagged array

```
System.out.println("Jagged Array:");
for (int[] row : jagged) {
    for (int num : row) {
        System.out.print(num + " ");
    }
    System.out.println();
}
sc.close(); }
```